

This assignment will not be graded and is only for practice.

Level 1:

1) Match the matrices in Column A with their inverses in Column B.

1 point

	Column A		Column B
a)	$A_1 = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 2 & 0 \\ 3 & 3 & 3 \end{bmatrix}$	i)	Inverse does not exist
b)	$A_2 = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 2 \\ 0 & 0 & 3 \end{bmatrix}$	ii)	$B_2 = \begin{bmatrix} 1 & 0 & 0 \\ -1 & \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} & \frac{1}{3} \end{bmatrix}$
c)	$A_3 = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 5 & 7 \end{bmatrix}$	iii)	$B_3 = \begin{bmatrix} 1 & -\frac{1}{2} & 0 \\ 0 & \frac{1}{2} & -\frac{1}{3} \\ 0 & 0 & \frac{1}{3} \end{bmatrix}$

Table: M2W2AQ1

☐ a → iii

☐ a → ii

☐ b → iii

☐ b → i

☐ c → i

☐ c → ii

No, the answer is incorrect.

Score: 0

Accepted Answers:

a → ii

b → iii

c → i

(Use the below information to answer questions 2 and 3).

Consider a system of linear equations

$$2x_1 - x_2 = 3$$

$$x_1 - x_3 = 3$$

$$x_2 - x_3 = 2$$

Let the matrix representation of the above system be $Ax = b$, where $A = \begin{bmatrix} 2 & -1 & 0 \\ 1 & 0 & -1 \\ 0 & 1 & -1 \end{bmatrix}$, $x =$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \text{ and } b = \begin{bmatrix} 3 \\ 3 \\ 2 \end{bmatrix}.$$

2) Choose the set of correct options.

1 point

☐ $A^{-1} = \begin{bmatrix} 2 & -1 & 0 \\ 1 & -1 & 0 \\ 1 & 0 & -1 \end{bmatrix}$

☐ $A^{-1} = \begin{bmatrix} 1 & -1 & 1 \\ 1 & -2 & 2 \\ 1 & -2 & 1 \end{bmatrix}$

☐ Adjoint of the matrix A is $\begin{bmatrix} 1 & -1 & 1 \\ 1 & -2 & 2 \\ 1 & -2 & 1 \end{bmatrix}$

☐ $\det(A) = 1$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$A^{-1} = \begin{bmatrix} 1 & -1 & 1 \\ 1 & -2 & 2 \\ 1 & -2 & 1 \end{bmatrix}$$

Adjoint of the matrix A is $\begin{bmatrix} 1 & -1 & 1 \\ 1 & -2 & 2 \\ 1 & -2 & 1 \end{bmatrix}$

$\det(A) = 1$.

3) Choose the set of correct options.

1 point

☐ $x_1 = -2$.

☐ $x_2 = 1$.

☐ $x_3 = -1$.

☐ None of the above.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$x_2 = 1.$$

$$x_3 = -1.$$

4) Choose the set of correct options.

1 point

- ☐ A system of linear equations $Ax = b$ is called a homogeneous system of linear equations if $b = 0$.
- ☐ A system of linear equations $Ax = b$ is called a non-homogeneous system of linear equations if $b = 0$.
- ☐ If v is a solution of the system of linear equations $Ax = b$, then $\frac{1}{c}v$ is a solution of system of linear equations $cAx = b$, where $c = 0$.
- ☐ Let $Ax = b$ be a system of linear equations. If A is invertible, then $\text{adj}(A)x = b$ also has a solution.

No, the answer is incorrect.

Score: 0

Accepted Answers:

A system of linear equations $Ax = b$ is called a non-homogeneous system of linear equations if $b \neq 0$.

If v is a solution of the system of linear equations $Ax = b$, then $\frac{1}{c}v$ is a solution of system of linear equations $cAx = b$, where $c \neq 0$.

Let $Ax = b$ be a system of linear equations. If A is invertible, then $\text{adj}(A)x = b$ also has a solution.

Suppose C is the cofactor matrix of $\begin{bmatrix} 3 & 2 & 3 \\ 1 & 0 & 1 \\ 2 & 1 & 1 \end{bmatrix}$. Let C_{ij} be the ij -th element of the cofactor matrix C . Answer the questions 5,6 and 7.

5) Find the value of C_{12} .

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 1

1 point

6) Find the value of C_{22} .

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) -3

1 point

7) Find the sum of the elements in the third row of C .

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 0

1 point

Level 2:

8) Which of the following square matrices of order 3 are the same as their adjoint matrices?

1 point

- ☐ Identity matrix.
- ☐ Zero matrix.
- ☐ Any scalar matrix.
- ☐ Any diagonal matrix.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Identity matrix.

Zero matrix.

9) Choose the set of correct options.

1 point

[Hint: For the first option, find out the adjoint matrix of an arbitrary square upper triangular matrix of order 2, i.e., take the matrix $\begin{bmatrix} a & b \\ 0 & d \end{bmatrix}$ and find out its adjoint matrix.]

- ☐ The adjoint of a 2×2 real upper triangular matrix is an upper triangular matrix.
- ☐ There exists a square matrix of order 3, such that $A = A^{-1} = \text{adj}(A)$.
- ☐ If $A = A^{-1}$, then $\det(A)$ must be 1.
- ☐ If $A = A^{-1}$, then A must be an identity matrix.
- ☐ If $A = A^{-1}$, then $A^2 = I$.
- ☐ If $A^{-1} = \text{adj}(A)$, then $\det(A)$ must be 1.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The adjoint of a 2×2 real upper triangular matrix is an upper triangular matrix.

There exists a square matrix of order 3, such that $A = A^{-1} = \text{adj}(A)$.

If $A = A^{-1}$, then $A^2 = I$.

If $A^{-1} = \text{adj}(A)$, then $\det(A)$ must be 1.

10) If A and B are square matrices of order 2, then choose the set of correct options. **1 point**

[Hint: Find out the adjoint matrix of an arbitrary square matrix of order 2, i.e., take the matrix

$\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and find out its adjoint matrix.]

- ☐ $\text{adj}(AB) = \text{adj}(A)\text{adj}(B)$.
- ☐ $\text{adj}(AB) = \text{adj}(B)\text{adj}(A)$.
- ☐ $\text{adj}(A + B) = \text{adj}(A) + \text{adj}(B)$.
- ☐ $\text{adj}(A^T) = \text{adj}(A)^T$.
- ☐ $\text{adj}(A^{-1}) = \text{adj}(A)^{-1}$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\text{adj}(AB) = \text{adj}(B)\text{adj}(A).$$

$$\text{adj}(A + B) = \text{adj}(A) + \text{adj}(B).$$

$$\text{adj}(A^T) = \text{adj}(A)^T.$$

$$\text{adj}(A^{-1}) = \text{adj}(A)^{-1}.$$

Your score is: 0/10